

AKASH KUMAR SINGH

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PROFESSIONAL SUMMARY

Motivated Computer Science & Engineering undergraduate specializing in data science, predictive modeling, and end-to-end data processing pipelines. Proficient in Python, SQL, exploratory data analysis (EDA), and machine learning classification algorithms using Scikit-Learn, Pandas, and NumPy. Hands-on experience analyzing large-scale datasets (32,500+ records), feature engineering, statistical modeling, and deploying AI solutions such as Retrieval-Augmented Generation (RAG). Strong foundation in core CS principles (DSA, DBMS, OS) with demonstrated analytical thinking and data-driven problem-solving abilities. Eager to leverage technical expertise to deliver actionable insights as a Graduate Engineer Trainee (GET) or Associate Data Scientist.

EDUCATION

Bhilai Institute Of Technology

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Raipur, Chhattisgarh

2023 – 2027 (Expected) — CGPA: 6.8

Vig English School

Senior Secondary Examination (Class XII)

Jamshedpur, Jharkhand

2023 — Percentage: 88.0%

Vig English School

Secondary School Examination (Class X)

Jamshedpur, Jharkhand

2021 — Percentage: 70.0%

TECHNICAL SKILLS

Programming Languages: Python

Core Computer Science Concepts: Object-Oriented Programming (OOP), Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN)

Data Structures & Algorithms Knowledge: Data Structures & Algorithms (DSA), Space & Time Complexity Analysis

Databases: SQL, Relational Database Management, Data Modeling

Machine Learning & Statistical Modeling: Predictive Modeling, Classification Algorithms, Regression Algorithms, Data Science Libraries, Hyperparameter Tuning, Feature Engineering

Generative AI & Natural Language Processing (NLP): Retrieval-Augmented Generation (RAG), LLM Integration & Local Inference, Audio & Speech Processing

Web / Application Development: FastAPI, Flask, Role-Based Access Control (RBAC), RESTful APIs, HTML, CSS

Tools & Platforms: Advanced Excel (Pivot Tables, Power Query, Power Pivot), Postman, Jupyter Notebook, Google Colab, FFmpeg, VS Code

Cloud / DevOps Exposure: Local LLM Inference Engines (Ollama), Streaming Pipeline Frameworks (Pathway)

Testing / Version Control: Git, GitHub, API Testing

Soft Skills: Analytical Problem Solving, Continuous Learning, Technical Documentation, Logical Thinking, Visual Communication

INTERNSHIPS / INDUSTRIAL TRAINING / TECHNICAL EXPERIENCE

RAYS IT & Design World Pvt. Ltd.

Vocational Trainee – AI Powered Data Analytics with Python & Advanced Excel

Raipur, Chhattisgarh

June 2026 – July 2026

- Acquired hands-on experience in data cleaning, preprocessing, and Exploratory Data Analysis (EDA) using Python libraries including NumPy and Pandas.
- Utilized Advanced Excel, Power Query, Power Pivot, and Pivot Tables to model unstructured datasets and automate business analytics reporting workflows.
- Developed and executed a capstone project titled "Freemium Gaming Monetization & Player Retention", building interactive analytics dashboards to track key player behavior metrics.

RAYS IT & Design World Pvt. Ltd.

Vocational Trainee – Web Development with Python & Django

Raipur, Chhattisgarh

June 2025 – July 2025

- Gained practical experience in full-stack web application development leveraging Python, Django framework, and relational database management systems.
- Implemented Object-Relational Mapping (ORM) and RESTful API patterns to handle database operations and secure data routing.
- Constructed scalable backend models, views, and template structures following standard SDLC practices.

PROJECTS

Real-Time Air Quality Monitoring System

July 2026 – Present

Academic / Major Project — Python, Pathway, OpenAQ API, RAG



- Engineered a high-throughput, pure-Python streaming data pipeline to continuously ingest and process environmental metrics from external REST endpoints using the OpenAQ API.
- Architected a real-time data processing layer using Pathway, optimizing low-latency streaming performance and continuous computation without external web UI dependencies.
- Integrated a Retrieval-Augmented Generation (RAG) layer, enabling natural language querying and real-time contextual analysis over continuous streaming data.
- Applied software design practices to build an extensible, scalable backend pipeline using Python libraries.

Freemium Gaming Monetization & Player Retention

June 2026 – July 2026

Data Analytics Project — Python, Pandas, NumPy, Advanced Excel, Power Query



- Conducted comprehensive Exploratory Data Analysis (EDA) on player engagement datasets using Python (NumPy, Pandas) to identify churn triggers and in-game purchase patterns.
- Leveraged Advanced Excel, Power Query, Power Pivot, and interactive Pivot Tables to aggregate multi-source game telemetry data and automate reporting pipelines.
- Engineered structured business dashboards highlighting retention curves, cohort metrics, and player lifetime value (LTV) drivers to inform monetization strategies.

Census Income Predictive Modeling

Feb 2026 – Mar 2026

Data Science / ML Project — Python, Pandas, NumPy, Scikit-Learn, Seaborn



- Conducted exploratory data analysis (EDA) and data preprocessing on 32,500+ records from the Adult Census Income dataset using Pandas, NumPy, Seaborn, and Matplotlib.
- Implemented data cleaning pipelines involving duplicate removal, mode-based categorical imputation, label encoding, and feature scaling via Scikit-Learn.
- Trained, evaluated, and benchmarked 5 supervised machine learning models (including Random Forest, Support Vector Machines, and Logistic Regression) to classify income tiers.
- Optimized predictive capabilities, selecting Random Forest as the optimal algorithm based on an evaluation metrics achieving 85.3% accuracy and a 0.898 ROC-AUC score.

Generative AI Course Context Engine

Aug 2025 – Oct 2025

Academic / Personal Project — Python, FastAPI, Ollama, Whisper, Scikit-Learn



- Developed a local Retrieval-Augmented Generation (RAG) backend to process educational video assets using FFmpeg for audio extraction and OpenAI Whisper for automated transcription.
- Designed an offline embedding and vector retrieval workflow using Ollama (bge-m3 model), Pandas, Joblib, and Scikit-Learn cosine similarity metrics for semantic query matching.
- Deployed modular FastAPI REST endpoints integrated with llama3.2 to deliver context-aware Q&A, content summarization, and automated JSON-formatted quiz generation.
- Handled end-to-end backend integration, structured JSON data responses, and local LLM execution environments.

ADDITIONAL INFORMATION

- **Certifications:** CodeWithHarry – The Ultimate Job Ready Data Science Course — CodeWithHarry – Complete 2025 Python Bootcamp
- **Coding Platforms Link:** GitHub profile (github.com/razesoni) showcasing open-source backend, machine learning, and data analytics projects.
- **Achievements / Co-Curricular Activities:** Expressive sketching and illustration focused on structural layout and visual communication; continuous learner of deep narrative literature.